

Back To Practice

< Q1 >

Save

LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

Select Language : C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, frames;
5
6     printf("Enter number of pages in reference string: ");
7     scanf("%d", &n);
8     printf("\n");
9
10    int pages[n];
11
12    printf("Enter the reference string: ");
13    for(int i = 0; i < n; i++) {
14        scanf("%d", &pages[i]);
15    }
16    printf("\n");
17
18    printf("Enter number of frames: ");
19    scanf("%d", &frames);
20    printf("\n");
21
22    int frame[frames], time[frames];
23    int faults = 0, hits = 0;
24
25    // Initialize
26    for(int i = 0; i < frames; i++) {
27        frame[i] = -1;
```

Test Cases 2

Run Sample Cases

Run All Cases

Back To Practice

< Q1 >

Save

LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

Completed

Select Language : C (GCC 9.2.0)

```
26  for(int i = 0; i < frames; i++) {
27      frame[i] = -1;
28      time[i] = -1;
29  }
30
31  int counter = 0;
32
33  for(int i = 0; i < n; i++) {
34      int found = -1;
35
36      // Check if page is already in frame
37      for(int j = 0; j < frames; j++) {
38          if(frame[j] == pages[i]) {
39              found = j;
40              break;
41          }
42      }
43
44      if(found != -1) {
45          // HIT
46          hits++;
47          time[found] = counter++;
48      } else {
49          // FAULT
50          faults++;
51
52          int lru_index = 0;
```

Test Cases 2

Run Sample Cases

Run All Cases

Back To Practice

< Q1 >

Save

LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

Select Language : C (GCC 9.2.0)

```
44     if(found != -1) {
45         // HIT
46         hits++;
47         time[found] = counter++;
48     } else {
49         // FAULT
50         faults++;
51
52         int lru_index = 0;
53
54         // Find least recently used
55         for(int j = 1; j < frames; j++) {
56             if(time[j] < time[lru_index]) {
57                 lru_index = j;
58             }
59         }
60
61         frame[lru_index] = pages[i];
62         time[lru_index] = counter++;
63     }
64 }
65
66 printf("\nTotal Page Faults: %d\n", faults);
67 printf("Total Page Hits: %d\n", hits);
68
69 return 0;
70 }
```

Test Cases 2

Run Sample Cases

Run All Cases